Sport & Exercise Biomechanics I

Sport and Exercise Science

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Sport & Exercise Biomechanics I (A22541)

Short Title:	Sport & Exe Biomech 1		
Faculty	Science and Computing		
Department	Sport and Exercise Science		
Cluster	Sports Technology		
Author	RBOLGER		
Credits	5	**Level:**	Introductory

Description of Module / Aims

The aim of this module is to provide students with an introduction to the core elements of sport and exercise biomechanics. The module will use applied examples and demonstrations to highlight the relevance of these elements in exercise, physical activity and sport.

Programmes

	BSc (Hons) in Health and Exercise Science (`WD_T0011_X`)	1 / 2 / M
	BSc (Hons) in Nutrition and Exercise Science (`WD_T0010_X`)	1 / 2 / M
`SPOR-0104`	BSc (Hons) in Software Systems Development (`WD_KDEVP_B`)	2 / 4 / E
	BSc (Hons) in Sport and Exercise Science (`WD_T0012_X`)	1 / 2 / M
`SPOR-0104`	BSc in Software Systems Development (`WD_KCOMC_D`)	2 / 4 / E

Indicative Content

- Introduction to linear and angular kinematics.
- Introduction to linear and angular kinetics.
- Qualititative biomechanics (e.g. non numerical description of a movement based on direct observation by coaches and teachers).
- Quantitative biomechanics (e.g. numerical measures from data colected during a movement).
- Exemplars of biomechanical skill breakdown
- Introduction to equipment used by sports biomechanists to collect data (e.g. Motion analysis systems, force plates and jump mats, timing gates, and video)
- Fluid dynamics

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Outline the basic biomechanical concepts underpinning movement in sport.
2. Perform basic applied calculations relating to kinematic and kinetic quantities (e.g. velocity, acceleration, force).
3. Identify the key aspects and applications of equipment used during biomechanical testing.
4. Collect biomechanical data of a sports movement (e.g. speed, angular data, jump data).
5. Report on collected/sample biomechanical data.

Learning and Teaching Methods

- Lectures
- Practicals

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3
Continuous Assessment	50%	4,5
Lab Report	10%	4
Lab Report	40%	5

Assessment Criteria

Final exam:

Labs:

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	12	
Independent Learning	99	

Essential Material

- Blazevich, A. _Sports Biomechanics - The Basics: Optimising human performance._. London: A&C Black, 2007.

Supplementary Material

- Hamill, J. and K.M. Knutzen. _Biomechanical Basis of Human Movement_ . Maidenhead: McGraw-Hill, 2009.
- Zatsiorsky, V.M. _Biomechanics in Sport - Performance enhancement and injury prevention_. Baltimore : Lippincott Williams & Wilkins, 2008.